

SESSION 3: PRIVACY, SECURITY & DEPLOYMENT

By the end of this session, we should be able to:

- distinguish data locality, confidentiality, and formal privacy;
- explain the separate roles of differential privacy, secure aggregation, and encryption;
- identify privacy and integrity threats in federated systems;
- design deployment controls for heterogeneous, unreliable, and regulated environments.

Key idea: Federated learning keeps raw data local, but it does not automatically eliminate privacy leakage, attacks, or compliance obligations.



Training
collaboratively
doesn't
automatically
guarantee privacy

Threats

- Gradient/model inversion (reconstruct inputs)
- Membership inference (was a record in training?)
- Property inference and data poisoning/backdoors

Risk factors

- Small batches, overfitting, rich gradients, text/images

Defenses

- Secure aggregation + encryption in transit/at rest
- Differential privacy with clipping + noise
- Regularization, larger batches, early stopping; audit and red-team



Differential Privacy Mechanisms

- **Definition**
 - (ϵ, δ) -DP: output barely changes when one record is added/removed
- **Core ingredients**
 - Clip per-client update to bound sensitivity
 - Add calibrated Gaussian noise; track privacy with an accountant
- **In federated learning**
 - Do clipping + noise on device; combine with secure aggregation
 - Monitor spent ϵ, δ ; stop or adjust when budget nears limit
- **Practical tuning**
 - Set clip norm from quantiles; choose noise to hit target ϵ
 - Larger batches and fewer rounds reduce privacy cost

Secure Aggregation Protocols

Purpose: server learns only the sum of client updates—never any individual update

Approaches

- Masking with secret sharing (pairwise masks cancel in the sum; threshold recovery)
- Homomorphic encryption (server aggregates ciphertexts)

Practical notes

- Handle dropouts with threshold schemes and mask-recovery
- Combine with DP; encrypt in transit and at rest
- Validate with audits and fault-injection tests

